

Installation of Solar PV Systems and Electrical Services at		Tasks: Lifting, panel installation, general roof works, inverter mounting, communications systems, AC & DC cable running, electrical breaker and associated component installation, roof penetrations	
SWMS No:			
Person responsible for developing SWMS:	Operations Officer	Signature: <i>Varun Nair</i>	Date:
Person responsible for approving SWMS:	Operations Officer	Signature: <i>Varun Nair</i>	Date:
Person responsible for implementing SWMS:	All tradesman/sub-contractors	Signature:	Date:
Resources/Trades Involved: Trades Involved: Plant and Equipment Used: Maintenance Checks:	Solar panel installation, inverter mounting, communications and power cable laying, distribution board alteration Drills, hand tools, cable ties, hacksaw, angle grinder, wire cutters, multimeters, ladders, silicone guns, drop saws Routine/daily check done by Operator		
Applicable Environmental Health and Safety Legislation: • Work Safety Act 2008 (ACT) ACT • Work Safety Regulations 2009 • Occupation Health and Safety (Manual Handling) Regulation 1997			
Approved Codes Of Practice or Australian Standards Applicable to the Works: • AS 1657-1992 Fixed platforms, walkways, stairways and ladders – Design, construction and installation • AS/NZS5033 (Installation of photovoltaic arrays) • AS/NZS 3000:2007 Electrical installations • AS 4777.1-2005 (Grid connection of energy systems via inverters) • AS 2161 Occupational Protective Gloves – Selection, Use and Maintenance • ACT First Aid in the Workplace Code of Practice • ACT Work Cover Manual Handling Code of Practice • ACT Work Cover Safe Work on Roofs Code of Practice Part 1 • AS/NQ 1891.4 Industrial fall-arrest systems and devices • AS/NZS 4509.2 Stand-alone Power Systems • AS/NZS 1768 Lightning Protection • AS/NZS 1170.2 Structural design action – wind • AS/NZS 3008 Electrical Installations – Selection cables			

Safe Work Method Statement

Training Requirements – Initial and Ongoing

- On-going safety training by way of tool box talks
- Installer/Subcontractor Induction

Personal Protective Equipment Required for Task

- Collared shirt, protective gloves, eye protection, ear protection, safety harness and lanyard, first aid kit, high visibility clothing, sunscreen, sunglasses, hat

Risk Score Calculator

	Consequence				
Likelihood	Minor	Substantial	Serious	Very Serious	Disaster
Practically Impossible	1	1	1	2	2
Remotely Possible	1	1	2	2	2
Possible	1	2	2	2	2
Likely	2	2	2	3	3
Almost Certain	2	3	3	3	3

Likelihood/Consequence	Risk Class
The hazard has potential to: • Cause minor injury • Be contained within the site boundary	1
The hazard has potential to: • Temporarily disable or seriously injure • Cause minor damage to the structure • Breach the site boundary and pollute local environment	2
The hazard has potential to: • Permanently disable or kill • Cause major damage to the structure • Have significant impact on the surrounding population and environment	3

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	Job Step <i>Break the job down into steps</i>	Potential Hazards <i>Identify the hazards associated with each step. Examine each to find possibilities that could lead to an accident or adverse environmental impact</i>	Initial Risk	Controls <i>Using the previous two columns as a guide, decide what actions are necessary to eliminate or minimise the hazards that could lead to an accident, injury or occupational illness or environmental impact</i>	Revised Risk	Person Responsible
1	Arrival of delivery truck carrying solar array materials Delivery truck to be parked and loading/unloading to be done with utmost care.	• Accident involving moving vehicles • Accident involving pedestrians	2	• All operators are to wear high visibility clothing when truck is entering the driveway • Delivery truck is to park in an approved zone	1	All operators
2	Unloading of delivery material from truck to PSC warehouse Materials will be located on pallets and lowered onto ground level using a tailgate loader	• Accident involving moving tailgate loader • Strain injury arising from using pallet jack to transport pallets from within the truck to the tailgate loader • Property damage caused by falling materials • Equipment damage caused by falling materials and/or improper handling of vehicle	2	• All operators are to wear high visibility clothing when truck is entering the driveway • Delivery truck is to park in an approved zone • Only trained and licensed pallet jack operators may use a pallet jack • Overloaded pallets may require materials to be removed manually • In this case, operators must have read and understood the Manual Handling Code of Practice • Pallets are to be inspected prior to loading onto forklift to ensure all materials are securely fastened • Any frayed or loose packaging is to be supported by ratchet straps	1	All operators
3	Removing solar panels from pallets and cardboard boxes Box cutter to be used to remove plastic wrapping materials and ratchet straps holding the panels in place on the pallet	• Lifting injury • Slips, trips and falls • Cutting injury using box cutter	2	• All operators must have read and understood the Manual Handling Code of Practice • Ensure the warehouse floor is free from any obstacles and waste • All waste to be disposed of • Protective gloves to be worn when using box cutters to avoid risk of cuts or abrasions	1	All operators
4	Packing and securing solar array materials for transport to site Solar array materials are to be wrapped using plastic wrap to	• Lifting injury • Slips, trips and falls • Cuts and abrasions from cutting off plastic wrap	2	• All operators must have read and understood the Manual Handling Code of Practice • Ensure the warehouse floor is free from any obstacles and waste • All waste to be disposed of	1	All operators

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	secure them for transport. If necessary, materials will be further fastened by ratchet straps	• Impact injury from holding solar panels		• Protective gloves to be worn when cutting off plastic wrap to avoid risk of cuts or abrasions		
5	Pre-start talk to discuss SWMS and induction	• Communication failure • Site hazards • Traffic hazards	2	• All operators must have undergone Installer/Subcontractor Induction • Each step in SWMS must be communicated to all operators	1	Operations Officer
6	Inspection of work site for frost, ice or dew	• Slips, trips and falls	3	• Inspection of the work site to be undertaken at 7:30am to ensure any frost, ice or dew has sufficiently dried to commence work • Work must not commence under any circumstances on a frosty, icy or wet roof • All roof work to take place in accordance with ACT Work Cover Safe Work on Roofs Code of Practice	2	All operators
7	Inspection of roof site for safe access and egress	• Slips, trips and falls • Wind hazard • Fall injury from ladders	3	• Ensure that proposed working area is free from all obstacles and/or waste materials and that environment is exposed to safe working conditions • Work must not commence under any circumstances on a frosty, icy or wet roof • All roof work to take place in accordance with ACT Work Cover Safe Work on Roofs Code of Practice • Ladders only to be used as a means of access, not as a work platform • Only single items to be carried up ladders • No single ladder to provide lift greater than 7.5-9m • Correct ladder angle: 75 degrees (1 out of every 4 up) • Ladder must extend 1m above step off point unless another sufficient hand hold is available	2	All operators
8	Inspection of personal protective equipment	• Impact injuries from stationary or falling objects	2	• All operators to wear: ○ High visibility clothing ○ Protective gloves	1	All operators

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		Cuts and abrasions to hands, feet or other areas of body		- Collared shirt - Sunscreen		
9	Inspection of weather conditions	- Impact injury from falling objects caused by wet ropes, chains or cables - Slips, trips and falls	2	- Weather conditions to be examined to determine if the roof is appropriate for work - Inspection of the work site to be undertaken at 7:30am to ensure any frost, ice or dew has sufficiently dried to commence work - Work must not commence under any circumstances on a frosty, icy or wet roof	1	All operators
10	Hazard assessment	Hazard identification and risk assessment will be conducted and any items not mentioned in the SWMS will be dealt with. All documentation is to be read and signed	2	Standard Solar Procedure: No work is to commence until all risk assessment documents are read, understood and signed by each team member	1	All operators
11	Unload equipment	- Lifting injury - Accident involving moving vehicles	2	- All operators must have read and understood the Manual Handling Code of Practice - Observe site rules and speed limits - Park only where permitted/instructed - No phone calls whilst driving	1	All operators
12	Take equipment on site	- Lifting injury - Slips, trips and falls	2	- All operators must have read and understood the Manual Handling Code of Practice - Ensure the path is clear of obstacles and potential slip hazards - Protective gloves to be worn when handling materials to avoid risk of cuts or abrasions	1	All operators
13	All workers put on restraint harness and other personal protective equipment if required	- Fall injury from roof - Property damaged caused by fall from roof	2	- Protective gloves to be worn when working - Restraint harnesses to be worn at all times when completing work on roofs within 2m of the boundary or on a roof with a profile greater than 35 degrees where the fall is greater than 2m - Anchor points to be installed by a person who is competently trained in their use and has undertaken Working at Heights training	1	All operators
14	Moving solar panels to installation location	- Lifting injury - Slips, trips and falls - Wind hazard	2	All operators must have read and understood the Manual Handling Code of Practice	1	All operators

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				• All roof work to take place in accordance with ACT Work Cover Safe Work on Roofs Code of Practice		
15	Analysis of roof profile	• Slips, trips and falls	3	• If roof is at an angle greater than 35 degrees and fall is greater than 2m, an abseil access must be used by a competently trained person • Anchor points to be installed by a person who is competently trained in their use and has undertaken Working at Heights training	2	All operators
16	Inspection of roof for hazards	• Slips, trips and falls	3	• All roof work to take place in accordance with ACT Work Cover Safe Work on Roofs Code of Practice • Ensure that proposed working area is free from all obstacles and/or waste materials and that environment is exposed to safe working conditions • Work must not commence under any circumstances on a frosty, icy or wet roof	2	All operators
17	Analysis of areas requiring trenching	• Buried services • Public safety • Unstable ground • Plant safety • Deep excavation or unsafe ground • Overhead powerlines • Working near moving plant • Stockpiles to close to trench • Working near traffic • Undermining existing structures	2	• Locate all services • Ensure barricades and signs are in place • Visual check of site before start of work each day • Daily plant inspection attended • Check location, check voltage, check sufficient vertical clearance and if within 3metres use spotter and mark with Tiger Tails • Keep 5 metre clearance of moving plant, reverse warning buzzers are working • Keep stockpiles outside of zones of influence • Comply with Traffic Control Plans (held on site) • Shoring of excavation, underpinning as required, temporary stays to power poles	1	All operators
18	Transporting equipment onto roof	• Ladder hazard • Slips, trips and falls • Wind hazard	3	• Ladders only to be used as a means of access, not as a work platform • Only single items to be carried up ladders • No single ladder to provide lift greater than 7.5-9m	2	All operators

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				• Correct ladder angle: 75 degrees (1 out of every 4 up) • Ladder must extend 1m above step off point unless another sufficient hand hold is available		
19	Working on elevated work platform	• Slips, trips and falls • Fall injury	3	• All installers must have undertaken Working at Heights training • All operators to wear required personal protective equipment	2	All operators
20	Installation of solar panels	• Lifting injury • Slips, trips and falls • Wind hazard	3	• All operators must have read and understood the Manual Handling Code of Practice • Ensure that proposed working area is free from all obstacles and/or waste materials and that environment is exposed to safe working conditions • All roof work to take place in accordance with ACT Work Cover Safe Work on Roofs Code of Practice	2	All operators
21	Working on roofs, grinding tiles, penetrations	• Fall injury • Injury caused by something being dropped • Sunstroke	3	• Fall Arrestors to be worn at all times while within 2m of roof edge • Suitable footwear to be worn at all times to minimise risk of slipping • All roof work to take place in accordance with ACT Work Cover Safe Work on Roofs Code of Practice • No tools to be left unattended on roof • Owners/occupiers to be informed that workers are on the roof and that they must not enter/exit the house near the work area • All operators to wear required personal protective equipment	2	All operators
22	Working in confined spaces	• Electrical shock or electrocution • Body injury from moving equipment if inadvertently activated or from falling or slipping • Body injury from direct contact with corrosive materials		• All installers must have undertaken Confined Spaces training during employee induction process		All operators

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		• Suffocation, asphyxiation or poisoning • Fire or explosion • Scalding or burning • Hypothermia, drowning • Burial				
23	Working with chemicals	• Respiratory problems and/or skin corrosion caused by exposure to chemicals	2	• Refer to respective MSDS • All operators to wear required personal protective equipment	1	All operators
24	Drilling holes in concrete and brickwork	• Hearing damage • Eye injury caused by flying debris • Respiratory/eye irritation case by dust • Asbestosis	2	• All operators to wear hearing protection, dust mask and safety glasses when operating drilling machine • If the presence of asbestos is suspected or identified, cease all work immediately	1	All operators
25	Angle grinding aluminium	• Injury to hand/body	2	• New operators to be supervised by senior installer to ensure proper tool practices • All operators to wear hearing protection and safety glasses when operating the angle grinder	1	All operators
26	Laying pipe in trenches	• Slips, trips and falls • Trench collapse • Water seepage • Plant too close to trench causing collapse/serious injury • Manual handling injury • Noise • Fumes • Falling/dropping pipe • Finger jamming • Compressed air/water	2	• Ensure ladder or ramp access for depth greater than 1 metre • Ensure shoring or benching if greater than 1.5 metre in depth or unstable ground, ensure to monitor/observe trench walls • Ensure no one is in trench when sand is placed into trench, all persons to stand clear • Lift plant/equipment into/out of trench with appropriate lifting devices/manual handling techniques • All operators to wear required personal protective equipment • Ensure installers have undergone First Aid Training • Switch off machines when not in use, take regular "air" breaks if using machines for prolonged periods of time • Ensure pipes are secured to lifting device when positioning in trench	1	All operators

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				• Ensure all temporary pressure connections are securely fixed, stand clear before pressure testing commences		
27	Constructing and mounting the frame for the inverter if required Constructing the frame for each individual inverter may involve drilling or cutting aluminium or galvanised steel	• Electrocution • Lacerations • Eye injury caused by flying debris • Hearing damage	2	• All operators to wear hearing protection, dust mask and safety glasses when carrying out drilling or cutting • Testing and tagging • Visual inspection prior to use	1	All operators
28	Installation of cable trays and conduit	• Lifting injury • Cuts and abrasions • Lacerations	2	• All operators must have read and understood the Manual Handling Code of Practice • Protective gloves to be worn when installing cable trays and conduit	1	All operators
29	Installation of breaker boxes adjacent to each array of panels	• Slips, trips and falls • Lacerations • Eye injury caused by flying debris during drilling	2	• All operators must have read and understood the Manual Handling Code of Practice • One panel from each string to always be disconnected to ensure the circuit is not connected and cables are not live • All operators to wear hearing protection, dust mask and safety glasses when carrying out drilling or cutting	1	All operators
30	Working with DC electrical cables	• Electrocution • Fire hazard	3	• Prior to commencing work with DC cables, roof cavity or drawings for location of power cables to be inspected, so they can be avoided • Panels to be covered with a sheet while working with DC electrical cable • One panel from each string to always be disconnected to ensure the circuit is not connected and cables are not live • Separation of cable ends from electrical contacts to be maintained • All items to be Portable Appliance Tested (PAT) on a 6-monthly basis • Items to be visually inspected prior to use	2	All operators

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31	Connecting and running DC cables	• Electrocution • Trips, slips and falls • DC arcing causing fire	3	• One panel from each string to always be disconnected to ensure the circuit is not connected and cables are not live • Cable lays to be kept in cable tray and conduit	2	All operators
32	Installation of deck tight	• Lacerations from power tools • Damage to roof • Eye injury caused by flying debris during drilling	2	• Protective gloves to be worn when installing deck tight • All operators must have read and understood the Manual Handling Code of Practice	1	All operators
33	Mounting of DIN rail enclosure for DC breakers and surge arrestors	• Electrocution • DC arcing causing fire	3	• One panel from each string to always be disconnected to ensure the circuit is not connected and cables are not live • Roof breakers to be turned off	2	All operators
34	DC wiring of surge arrestors and DC breakers adjacent to inverter	• Electrocution • DC arcing causing fire	3	• One panel from each string to always be disconnected to ensure the circuit is not connected and cables are not live • Roof breakers to be turned off	2	All operators
35	Moving inverters to installation location	• Lifting injury • Slips, trips and falls • Wind hazard	2	• All operators must have read and understood the Manual Handling Code of Practice • Ensure the path is clear of obstacles and potential slip hazards • All roof work to take place in accordance with ACT Work Cover Safe Work on Roofs Code of Practice	1	All operators
36	Mounting inverters	• Lifting injury • Slips, trips and falls	2	• All operators must have read and understood the Manual Handling Code of Practice • Ensure the path is clear of obstacles and potential slip hazards	1	All operators
37	Installing battery storage units	• Lifting injury • Slips, trips and falls • Electrocution	2	• All operators must have read and understood the Manual Handling Code of Practice • Ensure the path is clear of obstacles and potential slip hazards • Ensure all connections are made and commissioning has taken place prior to switching on • Ensure all cables are wired correctly as per Standard Solar Procedure	1	All operators

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38	Switching off inverters To ensure there is no live electricity flowing through the inverter	• Electrocution	3	• Ensure inverter is switched off at least 30 minutes before work is to be completed	2	All operators
39	Removing covers from inverter	• Lifting injury • Electrocution	3	• Ensure inverter is switched off before cover is removed • All operators must have read and understood the Manual Handling Code of Practice	2	All operators
40	Connecting DC cable into inverters	• Electrocution • DC arcing causing fire	3	• One panel from each string to always be disconnected to ensure the circuit is not connected and cables are not live • Inverters to be isolated on the AC side	2	All operators
41	Draining existing hot water tank	• Burns • Slips, trips and falls	2	• All operators to wear required personal protective equipment • Ensure installers have undergone First Aid Training	1	All operators
42	Lifting evacuated tubes onto roof	• Ladder hazard • Slips, trips and falls • Wind hazard • Lifting injury	3	• All operators must have read and understood the Manual Handling Code of Practice • Ensure the path is clear of obstacles and potential slip hazards	2	All operators
43	Installing evacuated tubes and other new components	• Slips, trips and falls • Burns	2	• All operators to wear required personal protective equipment • Ensure the path is clear of obstacles and potential slip hazards	1	All operators
44	Connecting solar hot water system	• Electrocution • Burns	2	• All operators to wear required personal protective equipment • Ensure installers have undergone First Aid Training	1	All operators
45	Removing all rubbish and tools from work site	• Slips, trips and falls	1	• All rubbish to be placed in the bins/skip provided • Only toolboxes that can be carried in one hand may be transported manually • All other tools and remaining equipment are to remain in the designated storage area on the roof • When removing tools, 3 points of contact must be maintained with the ladder at all times	1	All operators

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46	Hazard assessment Hazard identification and risk assessment will be made and any items not mentioned in the SWMS will be dealt with. Checklist to be filled out. All documentation is to be read and signed.	Slips, trips and falls	2	Standard Solar Procedure: No work is to commence until all risk assessment documents are read, understood and signed by each team member	1	All operators
47	Emergency rescue procedures In case of incapacitation of abseil worker whilst in operation	Medical emergency whilst working	2	- Buddy system to be used with two workers to be working at all times - First aid kit to be present at all times	1	All operators